Calcurion Maths Worksheets

A-Level Further Mathematics

Topic: Further Mechanics 1 – Centres of Mass

SOLUTIONS

Full worked solutions follow standard Further Mechanics 1 methods.

Use conservation of momentum, Newton's experimental law, Hooke's law and EPE, circular motion equations $F = \frac{mv^2}{r}$, centre of mass formulae, resolution along/perpendicular to line of centres for oblique impacts, and the work-energy principle as appropriate.

Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless otherwise stated.